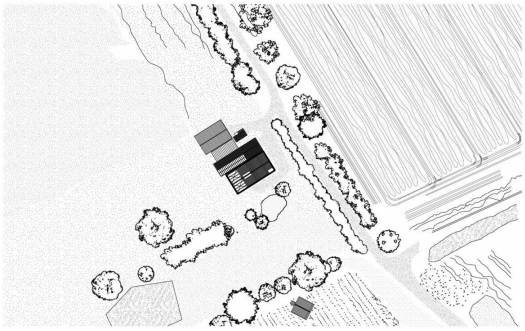
Arch504 - Hemp Flathouse

From farm to form: Flat house by Practice Architecture

Practice Architecture | Flat House

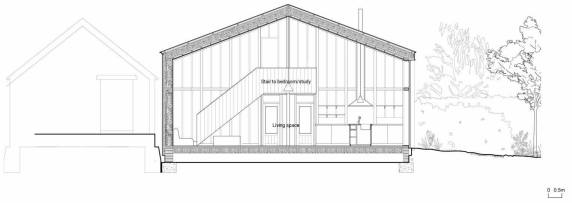
ARCH504 - Shared Drive

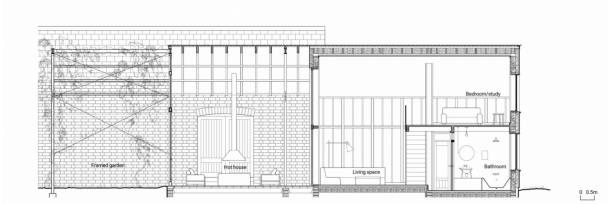
Site plan



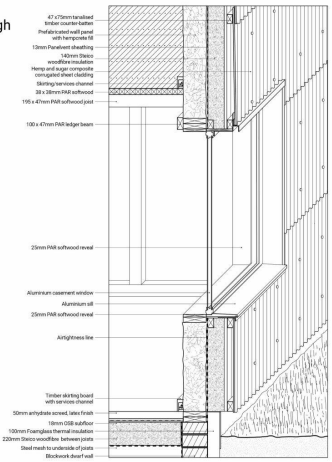


Section A-A





Perspective section through floor and wall



Questions to answer about the Hemp Flat House:

Where is it?

The Hemp Flat house is located in Margent Farms, Cambridgeshire, England

What is the environment like?

Rural, Farm, countryside

What are the surrounding buildings like?

The location is relatively isolated from surrounding buildings except for pre-existing agricultural sheds, yellow brick stores and open pole barn.

What are the traditional types of building there?

Buildings in surrounding towns are generally houses and shops made up of Brick, wood and concrete

How is it heated?

Heating and power is provided by a biomass boiler and a photovoltaic (PV) array – a system of solar-energy panels on the roof.

How is it ventilated?

The hemp house has passive ventilation from the windows, doorways and semi-open areas located within the site.

How is it waterproofed?

With the help of corrugated sheet cladding and a supportive groove junction between battens, It forms a rainscreen making the home weather resistant against rainy/stormy climate.

How is it insulated?

The Flat House contains wood fibre, foam glass as well as Hempcrete within the walls allowing healthy insulation to take place.

What is the structural system?

The building is made from a panelised system with scalability in mind. The structural system consists of timber I-joists with hempcrete infill and is based on the dimensions of a standard 8ft x 4ft sheet. Hempcrete, which is a mixture of hemp shiv and limestone, provides wall mass; it is both insulating and thermally massive. All of the shiv used in the house was grown on site and only the timber I-joists produced in the UK was used. There is also a layer of Woodfibre insulation that forms a woolly coat over the structure beneath the cladding. The building breathes, allowing moisture to pass through the wall freely and removing the need for any membranes.

How quickly did they build it?

Construction of the Flat House began in April 2018 and finished in March of 2019. 11 months altogether. The hempcrete panels are made and dried off-site which is why construction is able to be as fast as possible.

Can it be taken apart?

The hempcrete is constructed as boards that are built into the timber framing of the building so if needed, these can be deconstructed and used in other projects provided there is no damage to the boards.

How will they age?

Hempcrete is an incredibly strong material and is said to last 100’s of years so these boards will not need to be replaced unless unforeseen damage is done to them. The Boards are also fire resistant so this will also make their ability to age much better.

Where do you cook, eat, sleep, read, poop, party, wash, talk, tell stories? What are the technical/architectural requirements for this?
(Think acoustics, light, ventilation, smells, privacy, space, warmth, comfort)

Is there anything unusual about it? Why is it like that?

References:
https://www.architectsjournal.co.uk/buildings/from-farm-to-form-flat-house-by-practice-architecture-2